Epistemic Geometry of Descriptions:
Cover Stability as a Criterion for Observable Validity

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Abstract

Every description of a system presupposes that the chosen observable induces a stable, resolution-consistent structure of distinguishable states. This presupposition is rarely examined. We show that it can fail in a structured and detectable way, and that the geometry of that failure is itself informative.

We introduce the notion of *epistemic geometry*: the spatial structure of descriptive validity induced by an observable over a system's parameter space. A description is locally valid where the observable induces a perturbation-stable cover of the state space; it fails where that cover collapses. The failure modes are not random — they form structured regions whose shape, dimension, and position depend on the observable's operator structure relative to the system.

We formalize this as a necessary stability condition, derive the resulting bounded admissible resolution regime, and introduce directional cover heights as a parameter-resolved map of descriptive stability. The geometry is description-relative: different observables induce different epistemic geometries over the same system.

Two model systems demonstrate structurally distinct failure modes. In the Kuramoto synchronization model, cover analysis identifies both the known synchronization transition and a secondary instability band at $\kappa_{c2} \approx 1.55\,\sigma$ — without prior specification of either boundary. In the conservative pendulum, the analysis reveals the separatrix and additional observable-specific instability bands within a single dynamical regime, absent for alternative observables.

What is commonly reported as information loss is more precisely a collapse of epistemic geometry: the chosen description ceases to support stable distinctions, not because the system changes, but because the description fails. The proposed criterion provides a constructive diagnostic for this failure.

1 Introduction

Every description of a physical or complex system involves a choice: which observable to measure, and at what resolution. These choices are rarely treated as objects of study in themselves. They are regarded as methodological preliminaries, not as sources of structured failure. This paper argues otherwise.

We propose that an observable induces a *geometry* on the parameter space of a system — a spatial structure of regions where the description is valid and regions where it is not. This geometry is not an artefact of noise or numerical error. It reflects a structural relationship between the observable's operator composition and the system's state-space topology. The failure modes are organised, predictable in their shape, and informative about the system-description relationship.

We call this the *epistemic geometry* of a description. Its study is prior to classical information theory: before asking how much information a description carries, one must ask where that description is structurally capable of carrying any.

Several existing frameworks acknowledge that information is description-relative. Observable-dependent entropy [1] shows that thermodynamic entropy varies with the choice of observable, presupposing however that the chosen observable already sustains stable distinctions. Algorithmic complexity [2] measures description length relative to a universal machine, but does not address whether the described states are physically stable. Relational approaches to quantum mechanics [3] treat information as relative to an observer but provide no operational criterion for when such relativity yields reproducible distinctions. Maximum entropy inference [4] treats information as encoding constraints on a prior, but assumes those constraints are well-conditioned. Each framework presupposes that the chosen description is adequate; none asks when this presupposition is satisfied — and when it is not, what the geometry of that failure looks like.

The present work is also related to several concepts in dynamical systems and topology. *Generating partitions* and symbolic dynamics [5, 6] study when a coarse-graining of state space faithfully encodes the dynamics; our cover construction generalises this by dropping the requirement of a generating property and replacing it with a perturbation-stability criterion. *Coarse graining* in statistical mechanics [7] makes observable resolution explicit but typically presupposes that the chosen graining is adequate. *Observability* in control theory [8] asks when a system's state can be reconstructed from its outputs, which is complementary to our question of when output distinctions are themselves stable. *Persistent homology* [9] tracks topological features of data across resolution scales; our cover-height measure is conceptually related to persistence, in that both quantify how long a structural feature survives as ε varies, though we work directly with observable values rather than with the topology of an underlying point cloud. *Structural stability* [10] in dynamical systems asks when qualitative behaviour is preserved under perturbations; our stability criterion applies the same logic to the observable cover rather than to the flow itself.

This paper proposes a minimal criterion for the existence of descriptive validity. The core claim is that a stable observable state description exists only where the chosen observables and resolution jointly induce a non-trivial cover of the state space that remains stable under perturbations. This criterion does not replace classical information theory; it specifies a precondition for it. Shannon-type measures require distinguishable symbols. We ask when such distinguishability is structurally well-defined.

The contribution of the paper is threefold. First, we formalise epistemic geometry as a stability condition on observable-induced covers. Second, we derive the resulting bounded admissible resolution regime and introduce directional cover heights as a parameter-resolved map of descriptive validity. Third, we demonstrate on two model systems that the geometry of descriptive failure is structured, system-dependent, and recoverable without prior knowledge of either the failure locations or their causes.

We emphasise one distinction that is easily missed. The criterion does not measure how strongly an observable changes under perturbations — that is sensitivity analysis. It asks whether the observable induces stable, resolution-consistent distinctions at all. These are logically independent: large gradients may coexist with stable cover structure, and small gradients may fail to produce stable covers if the observable landscape has no persistent clustering geometry. The epistemic geometry of a description is not determined by its sensitivity alone.

The broader reframing is this: what is commonly reported as information loss, regime collapse, or descriptive failure is more precisely a collapse of epistemic geometry. The description ceases to support stable distinctions — not because the system changes, but because the description fails. The proposed criterion distinguishes these two cases constructively.

2 Observable Distinguishability

Let X be a state space and let

$$O : X \to \mathbb{R}$$

be an observable. Let $\varepsilon > 0$ denote a resolution threshold.

Definition 1 (ε-neighborhood)**.** *The ε-neighborhood of a state $x \in X$ with respect to observable O is*

$$U_\varepsilon(x) := \{x' \in X : |O(x') - O(x)| < \varepsilon\}.$$

Theoretical setting: pullback cover

For the purpose of formal analysis, we work with a *pullback cover* construction. Let $\mathcal{I} = \{I_j\}_{j=1}^k$ be a finite collection of open intervals of length ε that covers the observable image $O(X) \subseteq \mathbb{R}$. The pair $(O, \mathcal{I})$ induces a cover of X via pullbacks:

Definition 2 (Pullback cover)**.** *Given observable O and interval collection $\mathcal{I}$, the* pullback cover *of X is*

$$\mathcal{C}(O, \mathcal{I}) := \{O^{-1}(I_j)\}_{j=1}^k.$$

Each element $O^{-1}(I_j) = \{x \in X : O(x) \in I_j\}$ is the preimage of an interval of width ε.

This construction is well-defined for any measurable O and any finite open cover $\mathcal{I}$ of $O(X)$. It does not require connectivity of X and is unaffected by the path-concatenation issue that arises in adjacency-graph constructions on connected spaces (see Remark 2 below). The cover elements may overlap (when adjacent intervals overlap) and need not partition X; the collection $\mathcal{C}(O, \mathcal{I})$ is a genuine cover in the topological sense.

Remark 1 (Relation to Nerve and Čech constructions)**.** *The nerve of $\mathcal{C}(O, \mathcal{I})$ is a simplicial complex whose vertices are the intervals I_j and whose simplices record non-empty intersections of pullbacks. When $\mathcal{I}$ is a uniform cover with overlap $\delta < \varepsilon/2$, this nerve approximates the topology of $O(X)$ in the sense of the nerve lemma. The construction is thus a special case of the Mapper algorithm [11] applied to a scalar function, and relates to the Reeb graph of O when X is a manifold.*

Numerical setting: adjacency graph on a finite grid

In the numerical case studies (Sections 5 and 6), the state space X is approximated by a finite grid $X_h \subset X$ of n points at spacing h. All cover constructions operate on X_h. In this discrete setting, we use an adjacency-graph construction:

Definition 3 (Observable adjacency graph on X_h)**.** *Given observable O, resolution ε, and grid X_h, the* ε-adjacency graph *$G_\varepsilon(O, X_h)$ has vertex set X_h and an edge between $x, x' \in X_h$ whenever x, x' are grid-adjacent* and *$|O(x) - O(x')| < \varepsilon$.*

Definition 4 (Numerical observable cover)**.** *The* numerical observable cover *$\mathcal{C}_\varepsilon$ induced by (O, ε) on X_h is the set of connected components of $G_\varepsilon(O, X_h)$. The cardinality $|C|$ of a cover element $C \in \mathcal{C}_\varepsilon$ denotes the number of grid points in C.*

Remark 2 (Why adjacency-graph connectivity is well-defined on X_h)**.** *On a finite grid X_h, the adjacency-graph construction is free from the path-concatenation issue that afflicts the continuous case. In the continuous setting, a path through a level set of a smooth function can connect any two points with $|O(x) - O(x')| < \varepsilon$ via intermediate steps, collapsing the entire cover to a single component. On X_h, edges are only placed between* grid-adjacent *pairs, so the connectivity reflects local observable variation at scale h rather than global path-connectedness. The two constructions agree when O is monotone along grid paths; in general, the grid-adjacency construction is strictly finer. The pullback cover (Definition 2) provides the continuous-theory counterpart and is formally used in the proofs of Section 3 and Section 4.*

Remark 3 (Description-relativity of cover structure)**.** *The relevant object is not the raw state space X, but the cover structure induced on X by the pair (O, ε). State structure is description-relative: two descriptions (O, ε) and (O', ε') may induce entirely different covers of the same underlying space.*

A cover is *non-trivial* if it contains more than one component and each component contains more than a single grid point. In the numerical setting, singleton components arise as isolated grid points not connected to any neighbour under G_ε; they represent states that are not stably groupable at the given resolution, and are treated as degenerate. A cover that collapses to a single component makes no distinctions; a cover in which every component is a singleton makes no stable groupings. The admissible regime lies strictly between these extremes.

3 Perturbation Stability

Let Δ be a set of perturbations. The set Δ is left general; it may represent measurement noise, parameter uncertainty, or any class of variations that the description is required to be robust against. The choice of Δ defines the perturbation structure relative to which distinctions are evaluated, and thereby determines which observable neighborhoods are reproducible. Descriptive validity is therefore not an intrinsic property of the system, but exists only relative to a specified perturbation class Δ.

In the numerical case studies, Δ is instantiated as local adjacency in the discrete parameter grid: admissible perturbations are steps to nearest neighbours at grid spacing h. Cover stability therefore holds relative to this choice. Whether it persists under finer perturbation classes is a robustness question addressed empirically in Appendix A.

Definition 5 (Pointwise stability)**.** *A state $x \in X$ is* (Δ, ε)-stable *with respect to observable O if*

$$\sigma_\Delta(x) := \sup_{\delta \in \Delta} |O(x + \delta) - O(x)| < \varepsilon.$$

We call $\sigma_\Delta(x)$ the perturbation spread *at x.*

Pointwise stability requires that the observable value at x does not shift by more than ε under any perturbation in Δ. This is a worst-case criterion: if even one perturbation exceeds the resolution threshold, the state cannot be reliably located within a cover element.

Definition 6 (Stable cover element)**.** *A pullback cover element $O^{-1}(I_j) \in \mathcal{C}(O, \mathcal{I})$ is* Δ-stable *if every state $x \in O^{-1}(I_j)$ is (Δ, ε)-stable and satisfies*

$$\sigma_\Delta(x) < d(O(x), \partial I_j),$$

where $d(O(x), \partial I_j)$ denotes the distance from the observable value $O(x)$ to the boundary of the interval I_j. The cover $\mathcal{C}(O, \mathcal{I})$ is Δ-stable if every element is Δ-stable.

Remark 4 (Why distance-to-boundary is required)**.** *The condition $\sigma_\Delta(x) < \varepsilon$ alone does not guarantee that $x + \delta$ remains in the same cover element $O^{-1}(I_j)$. A state near the boundary of I_j may satisfy $|O(x + \delta) - O(x)| < \varepsilon$ while crossing into an adjacent interval under perturbation δ. The distance-to-boundary condition $\sigma_\Delta(x) < d(O(x), \partial I_j)$ is the correct sufficient condition: it ensures that the perturbed observable value $O(x+\delta)$ remains strictly inside I_j, so $x+\delta \in O^{-1}(I_j)$. In the numerical grid setting, this condition is approximated by $\sigma_\Delta(x) < \varepsilon - h\|\nabla O\|$, which accounts for the discretization error at scale h.*

Remark 5. *A cover element that fails the distance-to-boundary condition at some x is one where, under perturbation, the system appears to shift into an adjacent interval — making the distinction between elements unreproducible at that point.*

4 Descriptive Validity as Stable Cover Structure

4.1 The collapse threshold ε^*

Definition 7 (Cover-collapse threshold)**.** *Let $O : X_h \to \mathbb{R}$ and let $\mathcal{C}_\varepsilon$ denote the numerical cover of Definition 4. The* cover-collapse threshold *is*

$$\varepsilon^*(O, X_h) := \inf\{\varepsilon > 0 : |\mathcal{C}_\varepsilon| = 1\},$$

where $|\mathcal{C}_\varepsilon|$ denotes the number of connected components of the adjacency graph $G_\varepsilon(O, X_h)$.

Remark 6 (Properties of ε^*)**.** *(i)* Existence. *$\varepsilon^*(O, X_h)$ is the infimum of a non-empty set of positive reals (the cover becomes trivial at least at $\varepsilon = \operatorname{span}(O)$), so it exists and is finite.*

(ii) Uniqueness. *As the infimum of a set, ε^* is unique.*

(iii) Topology-dependence. *ε^* depends on the topology of the observable image $O(X_h)$ and the adjacency structure of X_h, not only on $\operatorname{span}(O)$. For a connected-interval image, $\varepsilon^* = \frac{1}{2}\operatorname{span}(O)$; for fragmented or non-interval images, ε^* can be substantially smaller. This is why the sufficient bound $\varepsilon < \frac{1}{2}\operatorname{span}(O)$ is not necessary in general.*

(iv) Cover-scheme independence. *ε^* is defined through the component count of $G_\varepsilon(O, X_h)$, not through a particular choice of covering intervals $\mathcal{I}$. It is therefore an intrinsic quantity of the pair (O, X_h).*

(v) Behaviour under refinement. *As the grid spacing $h \to 0$, $\varepsilon^*(O, X_h)$ converges to $\varepsilon^*(O, X)$, the collapse threshold of the continuous pullback cover. A formal convergence rate depends on the modulus of continuity of O; this is left for subsequent work.*

4.2 Descriptive validity

The following proposition records the immediate consequence of Definitions 2 and 6; it is logically direct but provides the formal anchor for the empirical analysis.

Definition 8 (Descriptive validity)**.** *Descriptive validity holds for the tuple $(X, O, \Delta, \varepsilon)$ if the pullback cover $\mathcal{C}(O, \mathcal{I})$ induced by (O, ε) is both non-trivial and Δ-stable in the sense of Definition 6. No ontological claim beyond Definitions 2 and 6 is intended.*

Proposition 1 (Consequence of cover instability)**.** *If no non-trivial Δ-stable cover exists for a given (O, Δ, ε), then descriptive validity fails at that descriptive level.*

Proof. Suppose the induced pullback cover $\mathcal{C}(O, \mathcal{I})$ is trivial. Then either all states fall into a single preimage $O^{-1}(I_j)$ for some j — meaning O makes no distinction across X at this resolution — or every preimage contains at most one point, in which case no stable grouping exists. In neither case can a state description be defined.

Suppose instead that $\mathcal{C}(O, \mathcal{I})$ is non-trivial but not Δ-stable. Then there exists a cover element $O^{-1}(I_j)$ and a state $x \in O^{-1}(I_j)$ such that $\sigma_\Delta(x) \geq d(O(x), \partial I_j)$. By Remark 4, there exists a perturbation $\delta \in \Delta$ such that $x + \delta \notin O^{-1}(I_j)$. Distinctions between cover elements are therefore not reproducible under perturbations in Δ. In both cases, descriptive validity fails at this descriptive level. □

This yields the admissible resolution regime: stability requires $\sigma_\Delta(x) < d(O(x), \partial I_j)$ for all x, and non-triviality requires $\varepsilon < \varepsilon^*(O, X)$, giving

$$\sup_x \sigma_\Delta(x) \;<\; \varepsilon \;<\; \varepsilon^*(O, X).$$

Corollary 1 (Lipschitz bound on perturbation spread)**.** *Let $O : X \to \mathbb{R}$ be Lipschitz continuous with constant $L > 0$, and let Δ be norm-bounded by r. Then $\sigma_\Delta(x) \leq L \cdot r$ for all $x \in X$, and the admissible resolution regime is non-empty whenever $L \cdot r < \varepsilon^*(O, X)$.*

Proof. $|O(x + \delta) - O(x)| \leq L\|\delta\| \leq Lr$ for all $\delta \in \Delta$. The regime is non-empty iff $Lr < \varepsilon^*(O, X)$. □

4.3 Cover height as a functional of the merge tree

We now develop the formal foundations of the cover height introduced operationally above. The key result is that cover height is a canonical functional of the *merge tree* of the filtration on X_h, strictly finer than the 0-dimensional persistence diagram.

Definition 9 (Edge-weight filtration and merge tree)**.** *Let X_h be a finite grid with adjacency graph G (edges between grid-neighbouring pairs). Assign to each edge $e = (x, x') \in G$ the weight $w(e) := |O(x) - O(x')|$. The* edge-weight filtration *on G is the sequence of subgraphs G_ε obtained by including all edges with $w(e) \leq \varepsilon$.*

The merge tree *$\mathcal{M}(O, X_h)$ records the history of connected-component merges as ε increases: each node of $\mathcal{M}$ corresponds to a connected component that exists for some interval $(\varepsilon_{\text{born}}, \varepsilon_{\text{died}})$, and each edge records a merge event at the ε value at which two components become one.*

Remark 7. *The merge tree $\mathcal{M}(O, X_h)$ is precisely the* dendrogram *of single-linkage clustering applied to X_h with edge weights $w(e) = |O(x) - O(x')|$. It is computable in $O(n \log n)$ time via Kruskal's algorithm on the grid graph.*

Definition 10 (Cover height)**.** *Let X_h be a finite grid of n points. The* cover height *$h(x)$ of a grid point $x \in X_h$ is*

$$h(x) = \frac{1}{n-1} \int_{\varepsilon_{\min}}^{\varepsilon_{\max}} \left(|C_\varepsilon(x)| - 1\right) d\varepsilon,$$

where $C_\varepsilon(x)$ is the connected component of $G_\varepsilon(O, X_h)$ containing x, and $|C_\varepsilon(x)|$ denotes the number of grid points in that component. The factor $(n-1)^{-1}$ normalises $h(x)$ to the interval $[0,\, \varepsilon_{\max} - \varepsilon_{\min}]$.

Proposition 2 (Merge-tree representation of cover height)**.** *Let $x \in X_h$ and let $\varepsilon_{\min} = e_0 < e_1(x) \leq e_2(x) \leq \cdots \leq e_{k(x)}(x) \leq \varepsilon_{\max}$ be the sequence of merge-event ε-values affecting the component containing x, with s_j the component size in the interval $[e_{j-1}(x), e_j(x))$ (where $e_0 = \varepsilon_{\min}$, $e_{k(x)+1} = \varepsilon_{\max}$). Then*

$$h(x) = \frac{1}{n-1} \sum_{j=0}^{k(x)} (s_j - 1)\,(e_{j+1}(x) - e_j(x)).$$

In particular, $h(x)$ is a functional of the merge tree $\mathcal{M}(O, X_h)$, not of the 0-dimensional persistence diagram PD_0 alone.

Proof. The integrand $|C_\varepsilon(x)| - 1$ is the step function that takes value $s_j - 1$ on each interval $[e_j(x), e_{j+1}(x))$ by definition of the merge-event sequence. The integral reduces to the stated sum.

That $h(x)$ is not recoverable from PD_0 alone follows because PD_0 records only the merge ε-values e_j as birth-death pairs $(0, e_j)$, without recording the component sizes s_j at each merge. Two distinct merge trees with identical PD_0 but different size sequences (s_j) yield different values of $h(x)$. □

Corollary 2 (Lower bound via persistence lifetime)**.** *Let $\mathrm{pers}_0(x) := e_1(x) - \varepsilon_{\min}$ be the 0-dimensional persistence lifetime of the component containing x at its birth (the ε-width of the interval during which x is an isolated point). Then*

$$h(x) \;\geq\; \frac{\mathrm{pers}_0(x)}{n-1},$$

with equality if and only if $k(x) = 1$ and $s_1 = n$ (the component containing x merges exactly once, directly absorbing all remaining points simultaneously).

Proof. The sum in Proposition 2 contains the term $j = 0$: $(s_0 - 1)(e_1(x) - \varepsilon_{\min})$. Since $s_0 = 1$ initially this term is zero; but the component of x has size at least 1, so the contributions for $j \geq 1$ are non-negative. The first non-trivial term is $(s_1 - 1)(e_2(x) - e_1(x)) \geq 0$. The sum of all terms for $j \geq 1$ equals $h(x) - 0 \geq 0$. Meanwhile, $\mathrm{pers}_0(x)/(n-1) = (e_1(x) - \varepsilon_{\min}) \cdot 1/(n-1)$, which is the contribution of the isolated-point interval scaled by $1/(n-1)$. The inequality holds because the remaining terms are non-negative. Equality holds iff all terms for $j \geq 1$ vanish, which requires $s_j = 1$ for all $j \geq 1$ except the final merge where $s_{k(x)} = n$ with $e_{k(x)+1} = \varepsilon_{\max}$ contributing zero width — i.e. exactly one merge event taking x from isolation to the full grid. □

Remark 8 (Why cover height exceeds persistence lifetime)**.** *Corollary 2 shows that the persistence lifetime* $\mathrm{pers}_0(x)/(n-1)$ *is generically a loose lower bound. Cover height is larger because it accumulates contributions from all subsequent merge events: a point that quickly joins a large component (small* $e_1(x)$*, large* s_1*) has high cover height even though its persistence lifetime is short. This is the correct behaviour for measuring* descriptive depth*: what matters is not merely how isolated* x *is initially, but how persistently it remains embedded in a large, stable region across all resolution scales.*

Remark 9 (Relation to Mapper and persistent homology)**.** *The merge tree* $\mathcal{M}(O, X_h)$ *is the 0-dimensional analogue of the Reeb graph of* O *on* X_h*. Standard persistent homology [9, 12] encodes* $\mathcal{M}$ *into* PD_0*, losing the component-size information. The Mapper algorithm [11] recovers a richer summary by retaining cluster structure within preimages, but does not integrate stability under* Δ*. Cover height occupies a complementary position: it retains size information from* $\mathcal{M}$ *and adds the* Δ*-stability layer via the integration bounds* $[\varepsilon_{\min}, \varepsilon_{\max}]$*, where* $\varepsilon_{\min} = \sup_x \sigma_\Delta(x)$ *enforces stability and* $\varepsilon_{\max} = \varepsilon^*(O, X_h)$ *enforces non-triviality.*

4.4 Directional cover heights and admissible window

Definition 11 (Directional cover heights and admissible window)**.** *For a two-dimensional parameter space with coordinates* p_1, p_2*, the* directional cover heights $h_{p_1}(x)$ *and* $h_{p_2}(x)$ *are defined by restricting the adjacency graph to edges along the* p_1*- and* p_2*-directions respectively, yielding separate merge trees* $\mathcal{M}_{p_1}$ *and* $\mathcal{M}_{p_2}$*. The* admissible description window

$$W(x) = \hat{h}_{p_1}(x) \cdot \hat{h}_{p_2}(x),$$

where $\hat{h}_{p_i} = h_{p_i} / \max h_{p_i}$ *denotes normalised directional height, provides a parameter-resolved map of descriptive validity.* $W(x) = 0$ *wherever either directional height vanishes;* $W(x) = 1$ *only where both are maximal.*

5 Case Study: Kuramoto Synchronization

5.1 Setup

Consider the Kuramoto model of coupled phase oscillators,

$$\dot{\theta}_i = \omega_i + \frac{\kappa}{N} \sum_{j=1}^{N} \sin(\theta_j - \theta_i),$$

with coupling strength κ, frequency dispersion σ (standard deviation of the Gaussian distribution of intrinsic frequencies ω_i), and standard order parameter

$$r\, e^{i\psi} = \frac{1}{N} \sum_{j=1}^{N} e^{i\theta_j}.$$

We take $O = r_{ss}$, the steady-state order parameter, and study its cover structure over the two-dimensional parameter space (κ, σ).

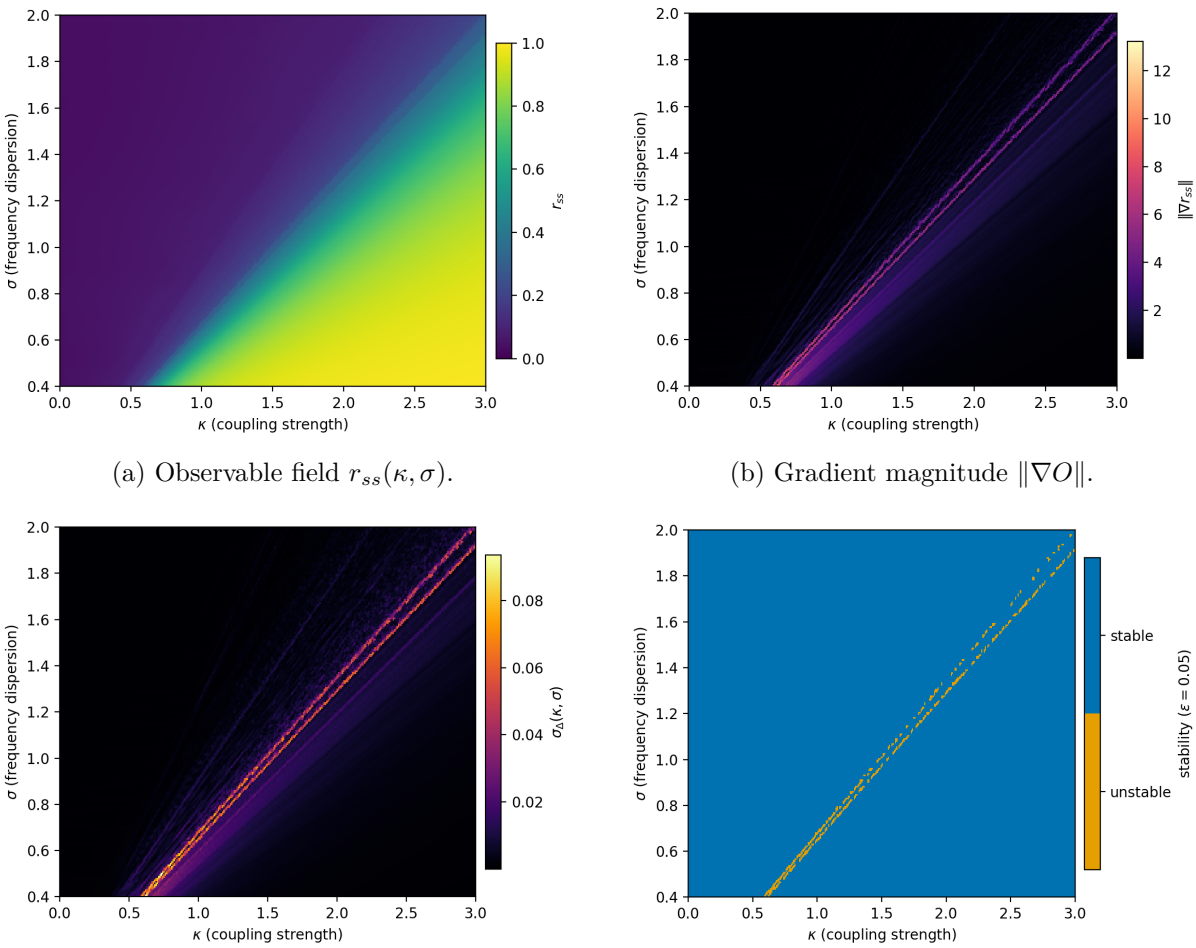
5.2 Descriptive collapse near the transition

Figure 1(a) shows the observable field $r_{ss}(\kappa, \sigma)$. A diagonal transition separates the incoherent regime (low r_{ss}) from the synchronized regime (high r_{ss}). The transition region is the domain of interest.

Figure 1(b) shows the gradient magnitude $\|\nabla O\|$: large values indicate where small parameter perturbations induce large observable shifts, directly measuring the risk of stability violation.

Figure 1(c) confirms this via the perturbation spread $\sigma_\Delta(\kappa, \sigma)$, computed as the maximum observable deviation under local parameter perturbations. High values cluster along the transition band, preventing stable observable neighborhoods precisely there.

Figure 1(d) makes the criterion explicit via a binary stability mask at $\varepsilon = 0.05$. The unstable region forms a structured band rather than a sharp boundary, shifting with σ and confirming that the stability condition depends jointly on coupling strength and frequency dispersion. The observable r_{ss} fails to induce a perturbation-stable cover precisely in the parameter region where the system undergoes its transition — consistent with Proposition 1.



(a) Observable field $r_{ss}(\kappa, \sigma)$.

(b) Gradient magnitude $\|\nabla O\|$.

(c) Perturbation spread σ_Δ.

(d) Stability mask ($\varepsilon = 0.05$).

Figure 1: Local stability structure of the Kuramoto observable r_{ss} over (κ, σ). **(a)** The observable field shows a diagonal transition separating incoherent and synchronized regimes. **(b)** The gradient magnitude identifies regions where small parameter perturbations induce large observable shifts. **(c)** The perturbation spread $\sigma_\Delta(x)$ confirms this: high values cluster along the transition band. **(d)** The binary stability mask at $\varepsilon = 0.05$ shows the unstable region as a structured band — consistent with Proposition 1.

5.3 Multiscale cover structure and secondary instability band

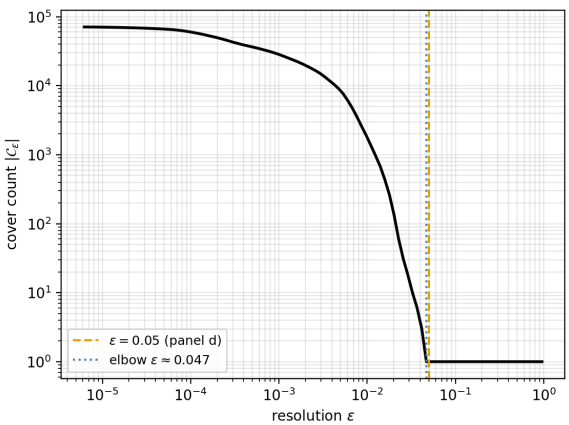
Figure 2(e) shows the cover count $|\mathcal{C}_\varepsilon|$ as a function of ε. As resolution coarsens, adjacent observable values merge and cover count decreases monotonically. The shape of this decay characterizes the multiscale distinguishability structure of the observable.

Figure 2(f) shows the observable-space cover height $h(x)$ from Definition 10. States that remain embedded in large covers over a wide range of ε receive high cover-height values, providing a multiscale measure of structural stability beyond local perturbation robustness.

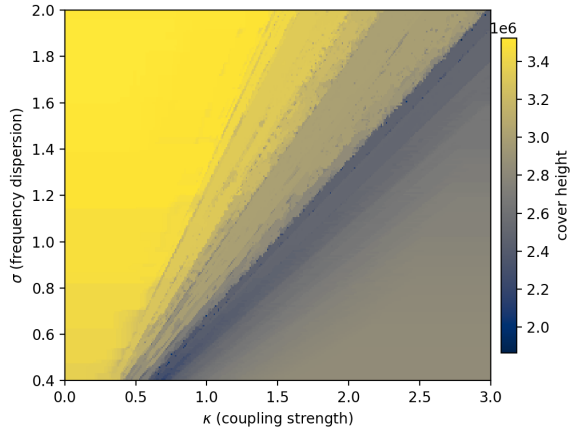
Figure 2(g) overlays the observable field with cover-height contours. Regions of high gradient coincide with reduced cover persistence, marking the collapse of stable structure near the transition boundary.

Beyond the primary transition, the cover analysis reveals a secondary instability band at $\kappa_{c2} \approx 1.55\,\sigma$ (at $N = 2000$; see Appendix B), visible as a second diagonal structure in the gradient magnitude and perturbation spread. This secondary band corresponds to a region of rapid variation in r_{ss} following the initial synchronization onset, where the observable undergoes a second phase of change before reaching its asymptotic value. What the cover method establishes is that cover stability is reduced there — the same criterion that identifies the primary transition identifies this structure without prior specification of either boundary.

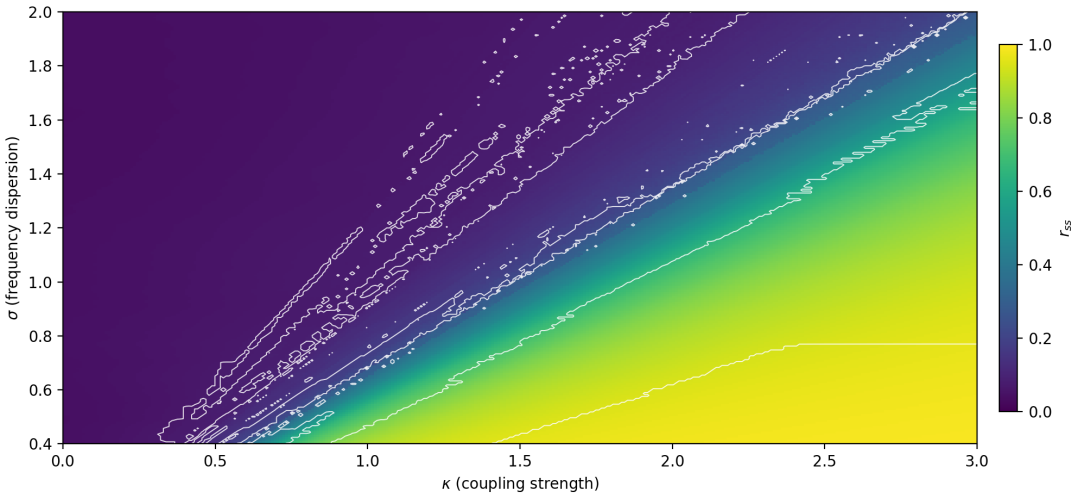
Together, these results show that descriptive validity — in the sense of Definition 8 — is not uniformly available across the parameter space, but emerges only in regions where local perturbation stability and multiscale cover persistence coexist.



(e) Cover count decay over ε.



(f) Observable-space cover height h.



(g) Observable field with cover-height contours.

Figure 2: Multiscale cover structure of the Kuramoto observable r_{ss} over (κ, σ). **(e)** Cover count decay over ε: as resolution coarsens, adjacent values merge monotonically. **(f)** Observable-space cover height assigns higher values to states embedded in large covers across a wide ε range. Cover-height contours track both the primary transition and the secondary instability band at $\kappa_{c2} \approx 1.55\,\sigma$ (at $N = 2000$; see Appendix B), recovering the full descriptive geometry without prior knowledge of either boundary. **(g)** Overlay of observable field and cover-height contours: high-gradient regions coincide with reduced cover persistence, marking the collapse of stable structure near both transition bands.

6 Case Study: Conservative Pendulum

The Kuramoto case demonstrates that the cover-stability criterion identifies the descriptive collapse near a statistical synchronization transition. To test the model-independence of the criterion, we apply the same construction to a structurally different system: a deterministic conservative pendulum with an analytic separatrix between bounded oscillation and unbounded rotation.

6.1 Setup

Consider the conservative pendulum

$$\ddot{\theta} + \omega_0^2 \sin\theta = 0,$$

with natural frequency ω_0 and conserved energy $E = \frac{1}{2}\dot{\theta}^2 - \omega_0^2 \cos\theta$. The separatrix energy is

$$E_{sep}(\omega_0) = 2\omega_0^2,$$

above which trajectories rotate rather than oscillate. We take the oscillation (or rotation) frequency $\omega(E, \omega_0)$ as the observable O, and study its cover structure over the parameter window $E \in [-1.8, 10.0]$, $\omega_0 \in [0.5, 2.0]$. The frequency is computed by direct numerical integration of the equation of motion. In contrast to Section 5, the system is fully deterministic: there is no ensemble average, no finite-N correction, and no random seed.

6.2 Descriptive collapse: separatrix and observable-specific bands

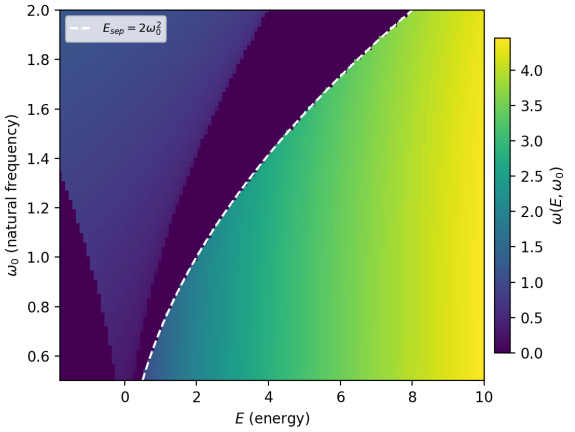
Figure 3(a) shows the observable field $\omega(E, \omega_0)$. The dominant feature is the sharp collapse of the frequency to zero at the separatrix $E_{sep} = 2\omega_0^2$, visible as a steep vertical drop. Figure 3(b) shows the gradient magnitude $\|\nabla\omega\|$: the largest values are concentrated along E_{sep} (overlaid in cyan), but a second, weaker ridge is also visible at $E \approx \omega_0^2$, roughly halfway to the separatrix.

Figure 3(c) shows the perturbation spread σ_Δ. The spread is large throughout the band $E \gtrsim \omega_0^2$, not only at the separatrix itself. Along $E = \omega_0^2$ the mean spread is approximately 0.36, far exceeding the resolution threshold $\varepsilon = 0.05$, so both ridges fail the stability condition.

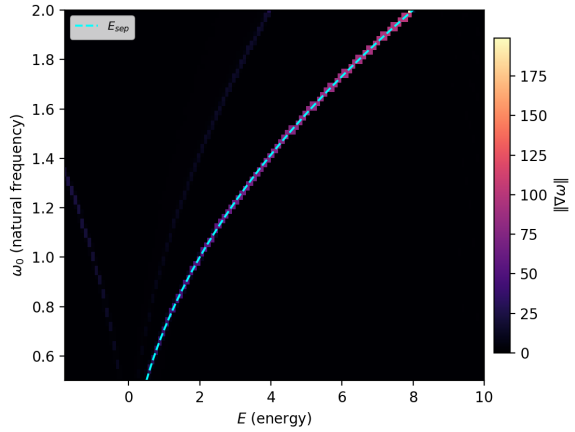
Figure 3(d) makes the criterion explicit. At $\varepsilon = 0.05$, the unstable region occupies approximately 4% of the parameter window, appearing as a band that encompasses both the separatrix and the secondary ridge.

The secondary ridge has a direct physical interpretation. The conservative pendulum frequency satisfies $\omega(E, \omega_0) \approx \omega_0\sqrt{1 - E/(2\omega_0^2)}$ near $E = 0$; the harmonic approximation loses $O(1)$ accuracy when $E \sim \omega_0^2$, i.e. when the energy reaches half the separatrix value. At this point $\partial\omega/\partial E$ increases sharply and the observable becomes sensitive to small energy perturbations.

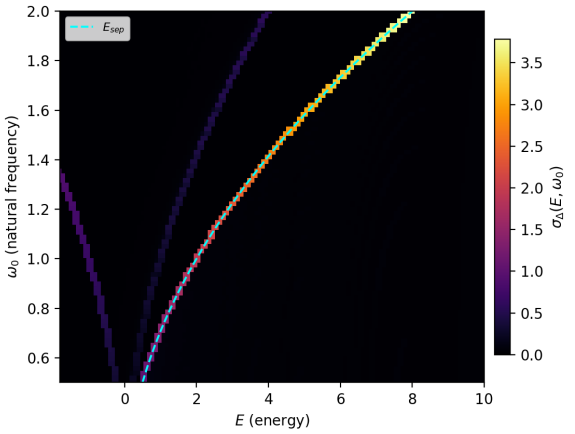
Crucially, this secondary band is a property of the *description*, not of the dynamics. The pendulum dynamics are smooth at $E \approx \omega_0^2$ — there is no phase transition there. A different observable — for example, oscillation amplitude — would produce a different cover structure and different instability bands. This directly instantiates Definition 8: the admissible cover depends on the tuple $(X, O, \Delta, \varepsilon)$, not on X alone. The cover-stability criterion identifies both the separatrix and this observable-specific crossover without any prior knowledge of the dynamical system.



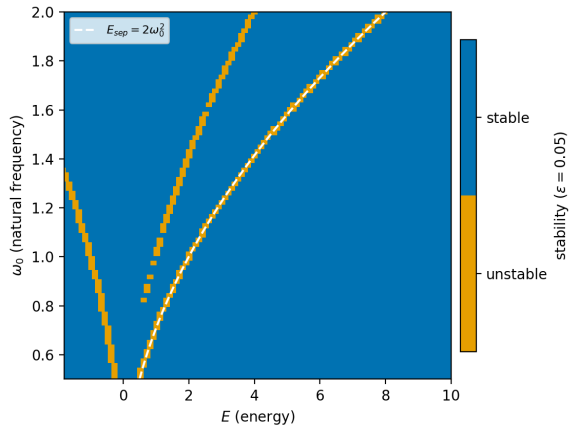
(a) Observable field $\omega(E, \omega_0)$.



(b) Gradient magnitude $\|\nabla\omega\|$.



(c) Perturbation spread σ_Δ.



(d) Stability mask ($\varepsilon = 0.05$).

Figure 3: Local stability structure of the conservative pendulum observable $\omega(E, \omega_0)$. **(a)** The frequency field exhibits a sharp collapse along the separatrix $E_{sep} = 2\omega_0^2$ (overlaid in white). **(b)** The gradient magnitude $\|\nabla\omega\|$ shows two ridges: a dominant one at E_{sep} and a secondary one at $E \approx \omega_0^2$ (the anharmonic crossover). **(c)** The perturbation spread σ_Δ is large throughout $E \gtrsim \omega_0^2$: both ridges exceed $\varepsilon = 0.05$. **(d)** The stability mask at $\varepsilon = 0.05$ identifies a connected unstable band encompassing both ridges.

6.3 Multiscale cover structure and directional heights

Figure 4(e) shows the admissible-window field, which quantifies how much resolution range a given parameter point can sustain a stable cover element across. Values are suppressed along both the separatrix and the secondary ridge at $E \approx \omega_0^2$.

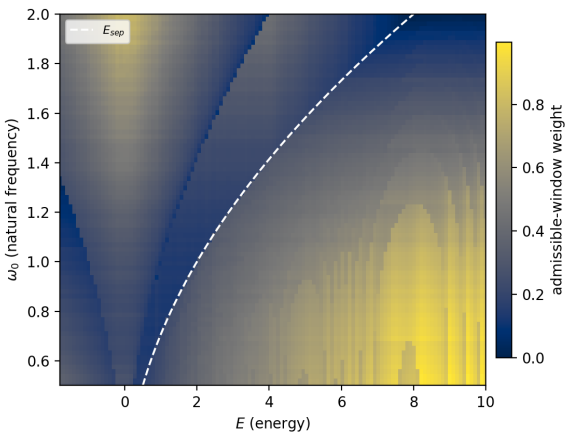
Figure 4(f) shows the observable-space cover height. The cover-height field is depressed in a band that begins at the anharmonic crossover $E \approx \omega_0^2$ and extends through the separatrix, broadening further on the rotation side where ω varies slowly with E.

Figure 4(g) overlays the observable field with both the cover-height contours and the analytic separatrix curve; the alignment is direct.

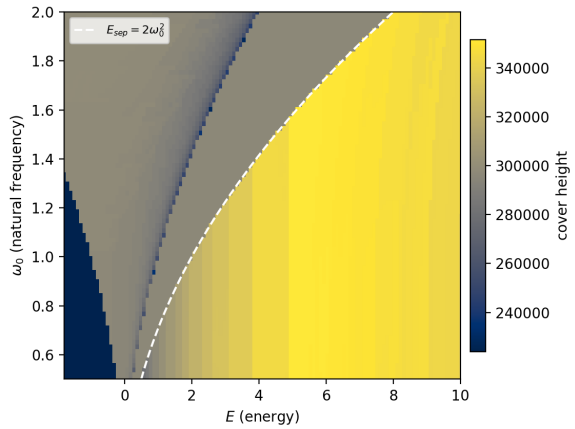
The directional cover heights h_E and h_{ω_0} differ substantially: h_E is primarily suppressed along the separatrix curve, while h_{ω_0} shows broader suppression across the oscillatory regime, reflecting the different sensitivity of ω to energy vs. natural-frequency perturbations. This anisotropy confirms that the admissible resolution interval is not isotropic in parameter space — consistent with Definition 11.

The conservative pendulum thus furnishes a case where the *ground-truth* stability boundary

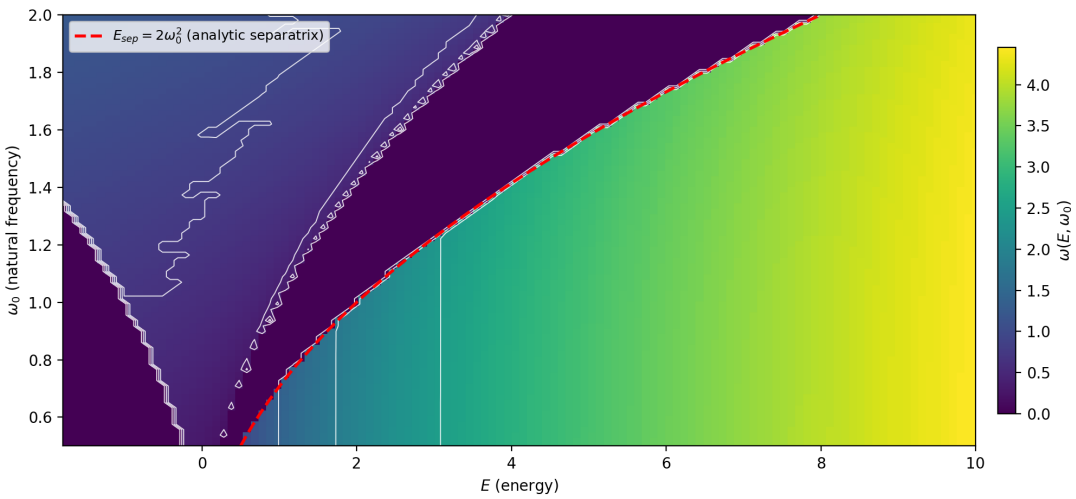
is known analytically. The cover formulation recovers it without making use of that knowledge, supporting the claim that the criterion of Definition 8 is genuinely model-independent.



(e) Admissible-window field.



(f) Observable-space cover height h.



(g) Observable field with cover-height contours and analytic separatrix $E_{sep} = 2\omega_0^2$ (red).

Figure 4: Multiscale cover structure of the conservative pendulum observable $\omega(E, \omega_0)$. **(e)** The admissible-window field is suppressed along both the separatrix and the secondary ridge at $E \approx \omega_0^2$. **(f)** Cover height is depressed from the anharmonic crossover through the separatrix, broadening on the rotation side where ω varies slowly. **(g)** Overlay of observable field, cover-height contours, and analytic separatrix (red): the cover-stability criterion recovers the separatrix from the observable alone, with no dynamical input — confirming model-independence of Definition 8.

6.4 Comparison with the Kuramoto case

Both case studies satisfy the same cover-stability criterion, yet they differ structurally:

- The Kuramoto case is statistical: r_{ss} is an ensemble average over N oscillators with random intrinsic frequencies, and the transition is a finite-size approximation to a phase transition in the $N \to \infty$ limit. The secondary band at $\kappa_{c2} \approx 1.68\,\sigma$ reflects a saturation feature of the observable field rather than a distinct dynamical phase boundary.
- The pendulum case is deterministic: $\omega(E, \omega_0)$ is a single-trajectory quantity, and the separatrix is an exact dynamical boundary. The secondary band at $E \approx \omega_0^2$ is an observable-specific sensitivity region with no dynamical counterpart — it would vanish under a different choice of O.

In both cases, the same cover construction — gradient, perturbation spread, stability mask, cover height — identifies the region where descriptive validity collapses. The geometry of the collapse differs: diagonal band in (κ, σ) for Kuramoto, parabolic curve in (E, ω_0) for the pendulum. Crucially, the Kuramoto secondary band is system-intrinsic (observable-driven but tied to a genuine saturation feature), while the pendulum secondary band is purely description-relative (absent for alternative observables). The criterion of Definition 8 treats both uniformly, which is the central methodological claim of this paper.

7 Discussion

The proposed criterion repositions information at a more fundamental level than classical information measures. Shannon entropy presupposes a symbol set. Here, the existence of a meaningful symbol set becomes the object of analysis.

This perspective has several consequences. First, it reframes information as a property of a description relative to a system, rather than of the system in isolation. Second, it distinguishes descriptive failure from physical disappearance: if an observable fails to induce a stable cover, information loss is local to that description. Third, it suggests that observable choice is not secondary but constitutive of information-bearing structure.

The two case studies illustrate this in complementary ways. In the Kuramoto model, descriptive failure occurs at a parametric phase boundary; the cover method recovers not only the known primary transition but also a secondary saturation boundary at $\kappa_{c2} \approx 1.55\,\sigma$ that is not visible in the observable field alone. In the pendulum, descriptive failure occurs at the separatrix — a topological boundary in phase space — and additionally at an observable-specific sensitivity region within the oscillatory regime. The secondary band in the pendulum is not a dynamical phase boundary; it reflects the local sensitivity structure of the frequency observable and would be absent under a different choice of O. Together, the two cases show that the cover criterion identifies both system-intrinsic and description-relative instability, and that the geometry of descriptive failure is itself structured and informative.

The cover formulation is deliberately weaker than a generating partition in the sense of symbolic dynamics [5, 6]: a generating partition requires that the coarse-graining faithfully encodes the full orbit structure of the dynamics, whereas we require only that cover elements are stable under a specified perturbation class Δ. A generating partition that fails perturbation stability would not yield descriptive validity in our sense; conversely, a stable cover need not be generating. The two conditions are therefore independent, and ours is weaker and more directly operational.

The relation to coarse graining in statistical mechanics [7] is similarly one of refinement rather than subsumption. Coarse graining fixes a partition and studies the resulting reduced dynamics; we instead ask which partitions are admissible given Δ and ε, making the choice of graining itself the object of analysis.

The cover-height measure introduced here is structurally related to persistence in topological data analysis [9, 12]: both track how long a feature survives as a resolution parameter varies. The difference is that persistent homology operates on the topology of a point cloud in feature space, whereas cover height operates directly on the scalar observable values and their stability under Δ. The two approaches are complementary and a formal connection remains to be developed.

The directional cover heights h_{p_1}, h_{p_2} and the admissible window $W = h_{p_1} \cdot h_{p_2}$ provide a parameter-resolved, continuous map of descriptive validity that goes beyond a binary stable/unstable classification. This quantifies how wide the admissible resolution interval is in each parameter direction, and where that interval collapses. The admissible window is therefore a constructive complement to the necessary condition of Proposition 1.

The relation to Mapper [11] and Reeb graphs deserves explicit comment. The pullback cover construction of Definition 2 is a scalar-function Mapper with a single covering interval of width ε.

Standard Mapper differs in two respects: it applies a clustering step within each preimage (here we use the whole preimage as a single cover element), and it is typically used for data-driven topological inference rather than stability analysis. The stability condition of Definition 6 — which requires that perturbations do not push states across interval boundaries — has no direct counterpart in Mapper. Reeb graphs [13] record the topology of level sets of O and are insensitive to the perturbation structure Δ; our framework adds precisely this perturbation layer. The cover-height measure of Definition 10 is then a Δ-weighted persistence measure on the Reeb graph of O restricted to X_h.

The argument is intentionally minimal. It does not require a particular domain ontology, nor does it depend on one specific theory of measurement. It is therefore applicable, in principle, across physical, biological, and complex systems wherever state distinctions depend on observables and resolution.

The present criterion can be seen as a minimal structural layer for frameworks that treat description and observability as primary objects, providing a necessary precondition that any such framework must satisfy before more specific structural claims can be made. The geometry of descriptive failure — its shape, dimension, and dependence on the observable's operator structure — is the natural object of study for such frameworks.

8 Outlook

The present result opens several natural follow-up questions.

Geometry of descriptive failure. The secondary structure in the Kuramoto case and the observable-specific bands in the pendulum suggest that the geometry of descriptive failure is itself a structured object. A formal theory of instability boundaries — their dimension, codimension, and relation to bifurcations — remains to be developed. In particular, the observation that instability regions form structured bands rather than isolated points suggests a connection to the theory of smooth maps and critical values that deserves formal treatment.

Directional resolution structure. The directional cover heights h_{p_1}, h_{p_2} introduce an anisotropic resolution structure on parameter space. Formalising this as a metric or tensor, and connecting it to existing notions of Fisher information or sensitivity analysis, is a natural next step. A partial order on observables — O_1 descriptively dominates O_2 at x if $h_{O_1}(x) > h_{O_2}(x)$ uniformly in ε — follows directly and defines a structured notion of descriptive adequacy that may be useful for observable design.

Multi-observable completeness. If information depends on observable-induced covers, one may ask when a set of observables is *complete* relative to a system: that is, when the joint cover structure induced by multiple observables is sufficient to resolve all relevant distinctions. A single observable may lose descriptive adequacy in one region while another remains stable there. The conditions under which a collection of observables jointly sustains a stable, non-trivial cover across the full parameter space constitute a natural multi-observable extension of the present criterion. The admissible resolution regime then becomes a region in $\mathbb{R}^k$ over a k-observable tuple.

Multiscale descriptive depth. The cover-height measure provides a first step toward a multiscale theory of descriptive depth. Its formal properties — continuity, monotonicity under coarse-graining, relation to topological persistence [9, 12] — remain to be established. Establishing a precise correspondence would connect the present framework to a mature computational toolkit.

Observable equivalence. The pendulum case shows that different observables produce different instability geometries for the same system. This suggests a structured notion of observable equivalence: two observables are equivalent at a point if they induce the same admissible window there. The corresponding equivalence classes and their relation to physical symmetries are left for subsequent work.

Operator signatures and cover geometry. The cover height $h(x)$ introduced here admits a deeper interpretation that connects to the operator structure of the observable itself. An observable composed from projection, aggregation, and asymptotic limit operations — such as the steady-state order parameter r_{ss} — will exhibit cover collapse precisely where those base operations lose their pre-theoretic validity: where ergodicity fails, where the stationary measure becomes non-unique, where the attractor limit diverges. This is not an accidental empirical fact but a consequence of the observable's operator composition.

More precisely: every non-trivial observable is a composition of base operations (projections, expectations, quotients, scalings, limits) that each presuppose structural conditions on the state space and dynamics. These presuppositions constitute the *pre-scopal substrate* of the observable. Cover collapse occurs where the substrate is violated — and the geometry of that collapse is determined by which base operations fail, along which parameter directions, and with what failure extent (point-like, band-like, or region-like).

A systematic theory connecting observable operator compositions to cover geometries — including formal characterisation of the dimension and codimension of the exclusion zone as a function of the substrate violation structure — would provide a predictive foundation for observable design: given the operator signature of a proposed observable, one could derive the geometry of its admissible description window before any simulation. This programme is left for subsequent work.

9 Conclusion

We have proposed a necessary condition for the existence of descriptive validity. Information exists only where the chosen observables and resolution jointly induce a non-trivial, stable cover of the state space under perturbations. This condition precedes classical information measures and clarifies when a system admits a meaningful descriptive alphabet at all.

The Kuramoto model and the conservative pendulum together illustrate that this condition is non-trivial and model-independent. In the Kuramoto case, the cover method identifies both the primary synchronization transition and a secondary saturation boundary at $\kappa_{c2} \approx 1.55\,\sigma$, neither specified a priori. In the pendulum case, the construction recovers the analytic separatrix $E_{sep} = 2\omega_0^2$ without dynamical input, and additionally identifies observable-specific instability bands within a single dynamical regime — bands that are absent for alternative observables and reflect the description-relative character of observable information.

Apparent information loss is therefore best interpreted, in at least some cases, as a failure of descriptive stability rather than as a disappearance of information itself. The admissible description window introduced here provides a constructive, parameter-resolved map of where stable description exists — and where it does not.

A Numerical Robustness of the Instability Ridge

The instability band reported in Section 5 was validated across three dimensions of numerical robustness. All robustness figures use the same observable r_{ss} and parameter window $(\kappa, \sigma) \in [0, 3] \times [0.4, 2.0]$ as Figure 1.

Resolution. Figure 5 compares the observable field at two grid resolutions (160×112 and 320×224). The band position and shape are stable across resolutions.

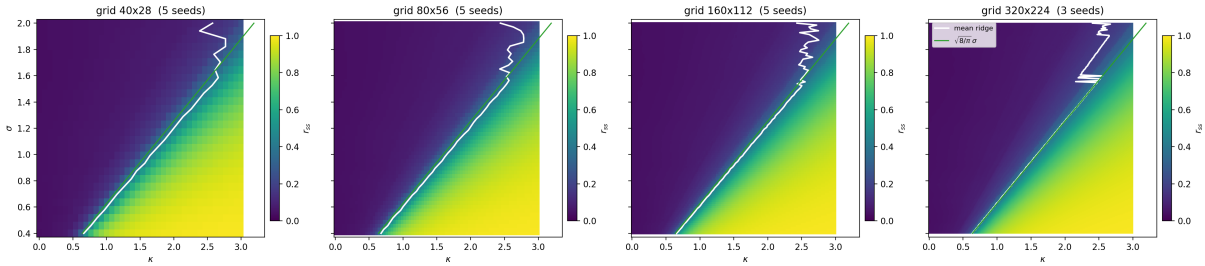


Figure 5: Mean r_{ss} field at two grid resolutions with instability ridge overlaid (white). Reference line $\sqrt{8/\pi}\,\sigma \approx 1.596\,\sigma$ is the analytic critical coupling for a Gaussian frequency distribution; the dashed line at $1.5\,\sigma$ shows the Lorentzian reference for context.

Seed and temporal convergence. Figure 6(a) shows the ridge position κ_c/σ for five independent random seeds at $N = 500$, $t = 200$. The grand mean is consistent with the analytic value $\sqrt{8/\pi}$ for a Gaussian frequency distribution; the seed-to-seed spread is small relative to the band width in Figure 1. Figure 6(b) shows the mean ridge position for $t \in \{50, 100, 150, 200\}$; the drift between $t = 100$ and $t = 200$ is below 1% of the ridge position.

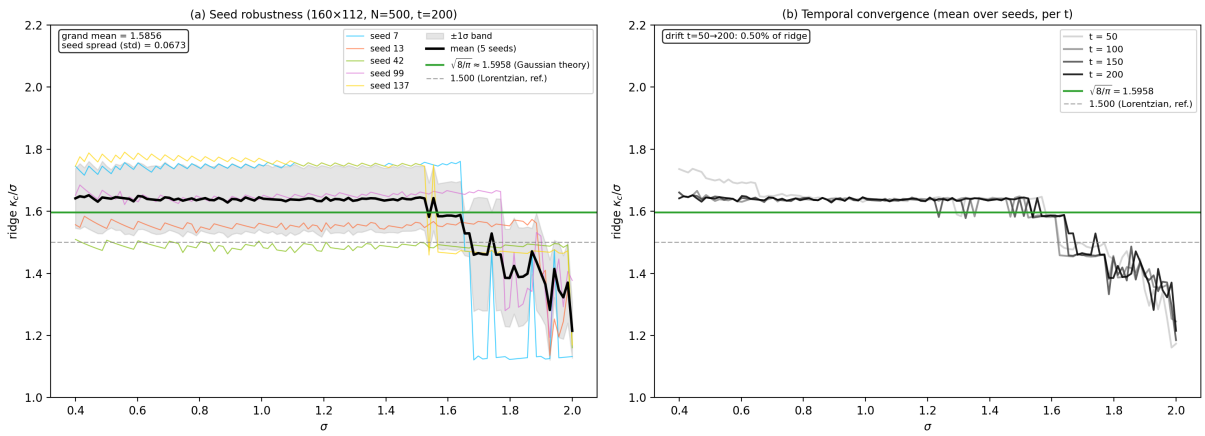


Figure 6: (a) Per-seed ridge position κ_c/σ at $N = 500$, $t = 200$, with $\pm 1\sigma$ band over five seeds. (b) Mean ridge position at $t \in \{50, 100, 150, 200\}$.

Oscillator count. Figure 7 shows the mean gradient field for $N \in \{125, 500, 2000\}$. The instability band remains present and similarly positioned across nearly two orders of magnitude in N.

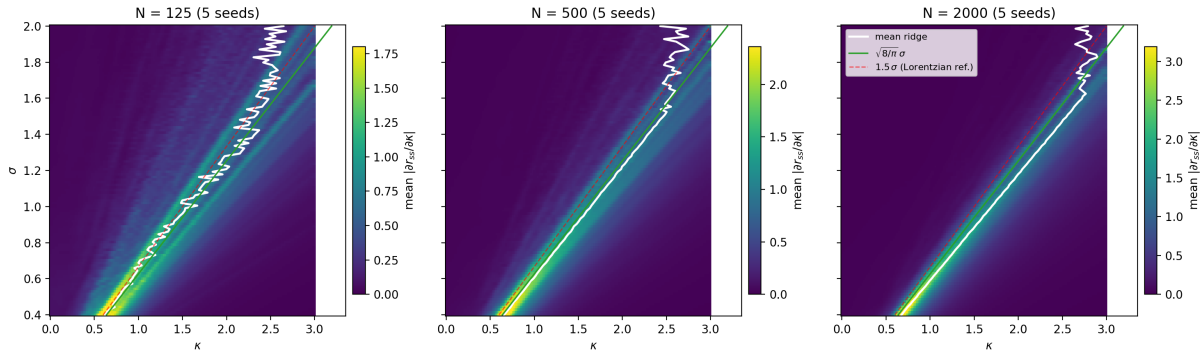


Figure 7: Mean gradient field $|\partial r_{ss}/\partial\kappa|$ for three oscillator counts N, with mean ridge overlaid (white).

Choice of frequency distribution. Intrinsic frequencies are drawn from $\mathcal{N}(0, \sigma^2)$. The theoretical critical coupling for a Gaussian distribution is $\kappa_c = \sqrt{8/\pi}\,\sigma \approx 1.596\,\sigma$, not the Lorentzian value $1.5\,\sigma$. The reference lines in Figures 5–7 reflect this. The distinction does not affect the cover-stability argument, which concerns the existence and structure of the instability band rather than its exact position.

B Robustness of the Secondary Instability Band

The secondary instability band reported in Section 5 was characterised across three robustness dimensions using 15 independent simulations ($N \in \{125, 500, 2000\}$, five seeds each).

Detection method. For each field $r_{ss}(\kappa, \sigma)$, the perturbation spread σ_Δ was computed via four-neighbour local maxima. Each σ-row was then mapped to the normalised coordinate κ/σ, and the row profiles were averaged to yield a one-dimensional band profile $\bar{\sigma}_\Delta(\kappa/\sigma)$. Both the primary band (κ_{c1}) and secondary band (κ_{c2}) were identified as local maxima of this profile via peak finding with minimum peak height 0.01 and minimum separation $\Delta(\kappa/\sigma) = 0.1$.

Results. Figure 8 summarises the findings. The normalised profile (top panel) shows two distinct peaks for all N at $N \geq 500$. Both peak positions and their separation are consistent across seeds at fixed N.

At $N = 2000$, the secondary band is located at $\kappa_{c2}/\sigma = 1.549 \pm 0.037$ (mean $\pm$ std over five seeds). The primary band lies at $\kappa_{c1}/\sigma = 1.688 \pm 0.057$, converging toward the analytic Gaussian prediction $\sqrt{8/\pi} \approx 1.596$ from above, consistent with the known finite-N upward shift [14]. The amplitude ratio of secondary to primary peak is 0.815 ± 0.067 at $N = 2000$, indicating that the secondary band carries substantial perturbation spread — it is not a weak residual.

ε-threshold dependence. The secondary band is visible in the continuous perturbation spread profile at all ε thresholds examined (0.03–0.05). It does not appear in the binary stability mask at $\varepsilon = 0.08$, because the band amplitude in σ_Δ is 0.035–0.050 — below that threshold. This is not evidence of artefact; it reflects that $\varepsilon = 0.08$ is too coarse to resolve the secondary structure. The admissible resolution regime for the secondary band requires $\varepsilon \lesssim 0.06$.

Finite-N behaviour. At $N = 125$ the two bands are not yet resolved: their peak positions are close and their seed-to-seed spread is large (std ≈ 0.3). At $N = 500$ the bands begin to separate; at $N = 2000$ the secondary band position is tight (std $= 0.037$, comparable to the primary band) and the profiles are consistent across all five seeds (bottom-right panel of Figure 8).

Interpretation. The secondary band at $\kappa_{c2}/\sigma \approx 1.55$ (at $N = 2000$) corresponds to a region where r_{ss} undergoes a second phase of rapid variation after the initial synchronization onset. The cover method identifies this as a second instability band without prior specification of its location. Whether this constitutes a distinct dynamical feature or a gradual saturation of the observable field is a question left for future analysis; the cover-stability criterion establishes only that descriptive validity is reduced there — and that this reduction is reproducible.

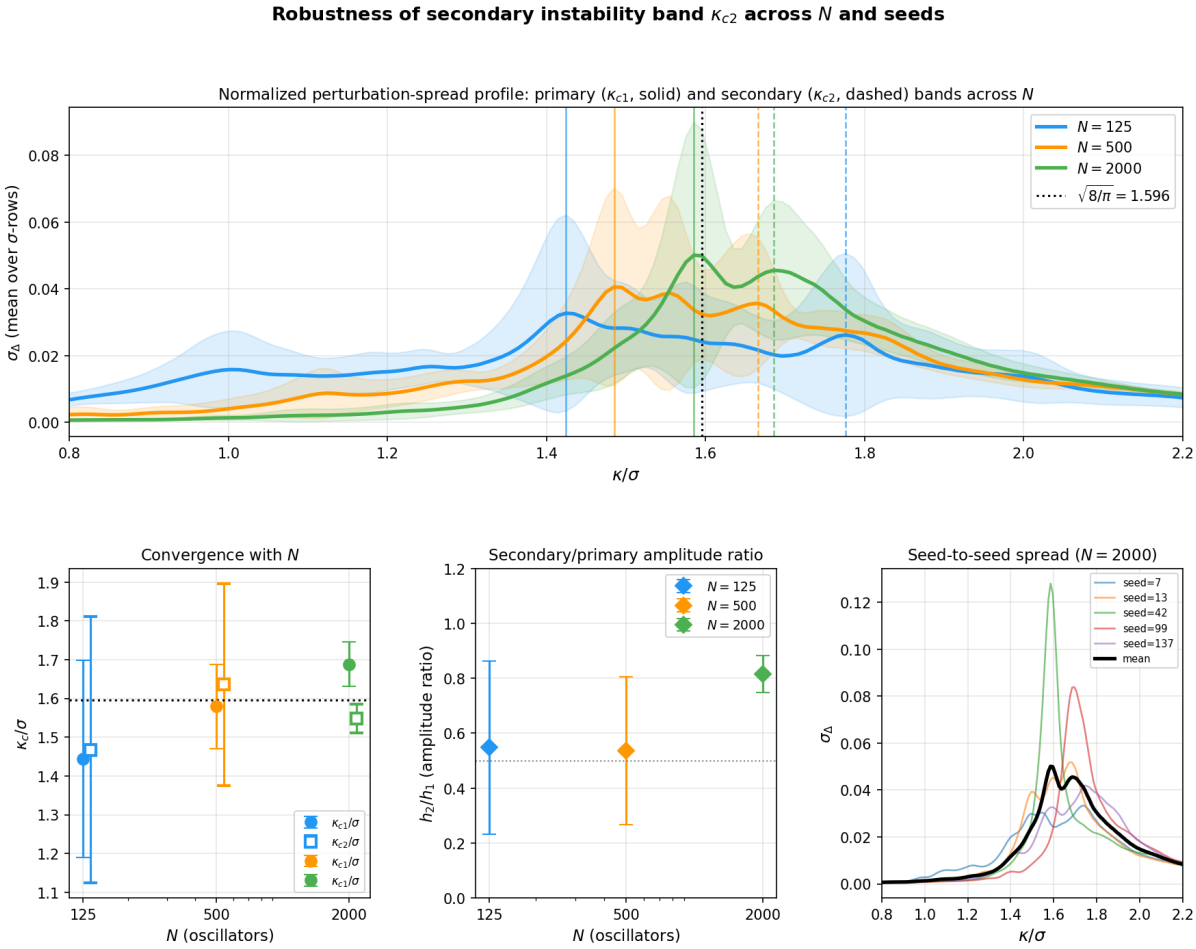


Figure 8: Robustness of the secondary instability band κ_{c2}. **Top:** Normalised perturbation-spread profile $\bar{\sigma}_\Delta(\kappa/\sigma)$ for $N \in \{125, 500, 2000\}$ (mean over five seeds; shaded band: $\pm 1\sigma$). Solid and dashed vertical lines mark κ_{c1} and κ_{c2} respectively; dotted line is the analytic prediction $\sqrt{8/\pi} \approx 1.596$. **Bottom left:** κ_{c1}/σ and κ_{c2}/σ vs. N (mean $\pm$ std over seeds). **Bottom centre:** Secondary-to-primary amplitude ratio h_2/h_1 vs. N. **Bottom right:** Per-seed profiles at $N = 2000$, showing tight seed-to-seed reproducibility of the secondary band.

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